Self-Regulated Photosynthesis Assistance System Using MXene-Based Multifunctional Sensors and Micro-LED Arrays

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Abstract
The emergence of an interconnected society has driven demand for wearable systems enabling seamless human-IoT integration. Two-dimensional (2D) materials have attracted significant interest due to their mechanical flexibility, lightweight nature, and metallic-level conductivity. However, practical applications face challenges including limited environmental stability and operational reliability. We present a multifunctional wearable platform utilizing 2D materials that addresses these limitations while enabling continuous real-time monitoring. By incorporating MXene (Ti3C2) as the functional layer, we developed a sensor capable of accurately detecting Voc, mechanical properties, and thermal parameters with wide-area responsiveness. Additionally, employing halogen ligand materials (FTP, BTP etc.) enables differentiation of sensing functional groups, allowing acquisition of detailed datasets. Individual components are integrated into a unified system supporting real-time, multi-modal signal tracking with enhanced specificity.
Keywords: MXene (Ti3C2), Multifunctional sensor, 2D Materials

1. Introduction

Accelerating climate change and population growth have created unprecedented demand for next-generation agricultural platforms that support efficient plant photosynthesis and optimized growth. Advanced cultivation systems such as vertical farms, plant factories, and smart greenhouses require integrated technologies that can precisely modulate light spectrum and intensity while providing real-time feedback on crop physiological status.

However, existing approaches face critical limitations including poor integration between light sources and sensing units, insufficient durability under harsh agricultural conditions (high humidity >85%, temperature fluctuations 25–60°C, continuous mechanical stress), and limited understanding of how prolonged LED exposure affects photosynthetic parameters. Additionally, the absence of real-time feedback mechanisms prevents dynamic optimization of light spectra, leading to energy waste and suboptimal growth outcomes. Conventional rigid sensing platforms cannot conform to irregular plant surfaces, limiting accurate localized measurements. These limitations underscore the urgent need for unified, miniaturized, flexible platforms capable of simultaneous photosynthetic enhancement and comprehensive real-time sensing in dynamic agricultural environments.

2. Result

To systematically address these interconnected challenges, we developed an innovative phytoelectronics platform based on heterogeneous integration of micro-light-emitting diodes (micro-LEDs) and MXene ($Ti_3C_2T_x$) multifunctional sensors. The carefully engineered device architecture features interdigitated MXene electrodes precisely patterned on elastomeric substrates composed of Ecoflex, a biocompatible silicone material known for its exceptional stretchability and durability. These electrodes are overlaid with high-density micro-LED arrays through an optimized soft lithography-assisted transfer printing process that ensures excellent optical coupling and mechanical integrity. This unique configuration enables intimate conformal attachment to irregular plant surfaces including leaves, stems, and even delicate growing shoots, while facilitating direct photon-sensor interaction that maximizes both illumination efficiency and sensing accuracy.

MXene, a two-dimensional transition metal carbide, serves as a multifunctional high-conductivity active layer with electrical conductivity reaching approximately 10^4S cm^{-1}, comparable to certain metals yet maintaining the mechanical flexibility of polymers. Beyond its excellent electrical properties, MXene imparts crucial functionalities including mechanical stretchability that accommodates plant growth and movement, chemical tunability through surface termination engineering, and strong optical absorption across the visible spectrum that enhances photothermal effects for temperature sensing. The integrated sensor demonstrates exceptional sensitivity to multiple parameters including open-circuit voltage (Voc) that correlates with local electrochemical gradients, temperature variations with sub-degree resolution, and mechanical strain with a remarkably broad linear response range spanning 0–30% deformation without degradation or hysteresis.

Rigorous mechanical and environmental testing revealed that the platform maintains stable, reproducible operation over 10^5 repeated bending cycles with minimal drift in sensor characteristics, and performs reliably under sustained high-humidity conditions (>85% relative humidity) for extended periods exceeding 30 days. These results demonstrate superior mechanical robustness and

environmental stability compared to conventional metal-oxide semiconductor sensors or conductive polymer-based devices, which typically fail within hundreds to thousands of cycles or degrade rapidly in humid environments. The exceptional durability stems from the intrinsic properties of MXene—including its layered structure that accommodates mechanical stress, hydrophilic surface terminations that prevent water-induced delamination, and chemical stability under physiological conditions.

Controlled illumination experiments conducted on model crop species including lettuce, strawberry, and tomato plants confirmed significant enhancement of chlorophyll content (15–25% increase) and measurably improved photosynthetic activity as quantified by CO_2 assimilation rates and quantum yield measurements under tailored red (630–660 nm) and blue (450–470 nm) micro-LED emission spectra optimized for photosynthetic absorption peaks. Simultaneously, the integrated MXene-based sensors provided continuous, high-temporal-resolution feedback on localized temperature fluctuations and surface potential variations that showed strong correlation with stomatal opening dynamics, transpiration rates, and overall plant water status. By correlating illumination conditions with physiological responses in real time, the platform enables identification of optimal light recipes for specific growth stages and environmental conditions.

Most significantly, the platform architecture allows dynamic, automated modulation of light intensity, spectral composition, and duty cycle based on real-time sensor feedback through a microcontroller-based control system. This closed-loop operation forms a "self-regulated" photosynthesis assistance system that adjusts illumination parameters to maintain optimal physiological conditions, minimize energy consumption, and prevent photo-oxidative stress from excessive light exposure.

3. Conclusion

This comprehensive study demonstrates that heterogeneously integrated MXene-micro-LED systems can successfully achieve high-performance multimodal sensing and efficient, targeted optical stimulation within a single compact, flexible architecture suitable for practical agricultural applications. The synergistic combination of exceptional electrical conductivity ($\approx 10^4$ S cm^{-1}), wide spectral compatibility spanning ultraviolet to near-infrared ranges, mechanical durability exceeding 10^5 cycles, environmental stability under agricultural conditions, and compatibility with scalable manufacturing processes firmly establishes MXene and related two-dimensional materials as ideal candidates for next-generation phytoelectronic interfaces and bio-integrated sensing platforms.

The proposed MXene-based phytoelectronics platform offers a reliable, cost-effective, and multifunctional solution that bridges fundamental materials science with applied agricultural biotechnology and precision farming practices. By enabling unprecedented integration of optical stimulation, physiological monitoring, and intelligent feedback control, this work establishes a new paradigm for sustainable and intelligent crop cultivation systems that can adapt dynamically to plant needs and environmental fluctuations. The platform's modular design allows straightforward customization for different crop species, growth stages, and cultivation environments, facilitating broad adoption across diverse agricultural sectors.

Looking forward, this research paves the way toward fully adaptive, data-driven smart farming systems capable of industrial-scale deployment in vertical farms, greenhouses, and even outdoor precision agriculture applications. By addressing critical challenges in global food security, resource efficiency, and climate-resilient agriculture through advanced materials integration and cyber-physical systems, our work contributes to the foundation of sustainable agricultural practices necessary to feed a growing global population while minimizing environmental impact. Future research directions include integration of artificial intelligence algorithms for predictive growth modeling, wireless sensor networks for large-scale deployment, and exploration of biodegradable substrate materials to further enhance the environmental sustainability of phytoelectronic platforms.

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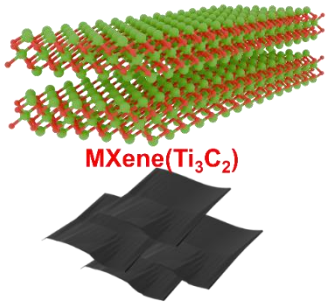

Fig. 1. 3D schematic of MXene